SS-3: Uniqueness of SU(3) from the Tetrahedral Cage

Conscious Point Physics — Strong Sector Series

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Abstract

SS-1 Theorem 1 proves that the eight DI-bit hopping operators on the tetrahedral cage base $\{V_1, V_2, V_3\}$ equal the Gell-Mann generators $T^a = \lambda^a/2$ and satisfy the SU(3) commutation relations exactly. That result establishes that SU(3) *can* emerge from the tetrahedral cage. This paper proves the converse: SU(3) *must* emerge. No alternative Lie algebra is consistent with the tetrahedral cage geometry.

The argument has three steps. First, the real vector space of traceless Hermitian operators on $\mathbb{C}^3$ has dimension exactly $3^2 - 1 = 8$. Second, this space equipped with the matrix commutator is, by definition, the Lie algebra $\mathfrak{su}(3)$. Third, the eight CPP operators are linearly independent (verified to machine precision, rank 8 Gram matrix), so they span the full space and generate $\mathfrak{su}(3)$ necessarily. The C_3 rotational symmetry of the cage selects the standard Gell-Mann basis within this unique algebra but does not alter the algebra itself.

Two corollaries follow: (i) the vertex count $N = 3$ determines the gauge group — $N = 2$ gives SU(2) (the electroweak sector), $N = 3$ gives SU(3) (the strong sector), and no other outcome is possible; (ii) no exotic gauge group (SO(8), Sp(4), G_2, etc.) can arise from three colour states.

Open Problem resolved: OPEN-SS-11 (SU(3) operator uniqueness).

Consequence: SS-1 Theorem 1 is elevated from a *possibility* result ("SU(3) is consistent with the cage") to a *necessity* result ("SU(3) is the unique algebra of the cage").

Keywords: SU(3), Lie algebra, uniqueness, tetrahedral cage, 600-cell, colour charge, Gell-Mann matrices, Conscious Point Physics, gauge group derivation, representation theory

Plain Language Summary: The strong nuclear force is governed by the symmetry group SU(3). In standard physics this symmetry is postulated. A previous CPP paper showed that the geometry of the tetrahedral quark cage *produces* SU(3). This paper proves it is the *only* symmetry the cage can produce. Three colour states and eight generators are not a choice — they are a geometric necessity.

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1 Introduction

The SU(3) colour gauge group is the foundation of quantum chromodynamics. In the Standard Model it is introduced as a postulate: quarks carry a three-valued quantum number (colour), and the gauge invariance of the colour degree of freedom generates the eight gluon fields.

In Conscious Point Physics (CPP), the three colour states are the three base vertices $\{V_1, V_2, V_3\}$ of the qCP tetrahedral cage embedded in the 600-cell lattice (Abshier and Grok, 2026a). SS-1 Theorem 1 (Abshier and Grok, 2026b) constructs eight DI-bit hopping operators T^a on these vertices and proves $T^a = \lambda^a/2$ exactly, with $[T^a, T^b] = if^{abc}T^c$. This is a constructive proof that SU(3) *can* emerge from the cage.

A natural question follows: is SU(3) the *only* algebra that can emerge? Could a different assignment of operators to the tetrahedral edges yield a different 8-dimensional Lie algebra — perhaps SO(8), Sp(4), or some exotic group — that is also consistent with the cage geometry?

If so, the SS-1 derivation would show that SU(3) is *accommodated* by the cage, not *required* by it. The result would be a possibility, not a prediction.

This paper answers the question: **no**. SU(3) is the unique Lie algebra of the tetrahedral cage. The proof is elementary linear algebra — it requires no representation theory beyond the definition of $\mathfrak{su}(N)$ and no computation beyond a rank check.

1.1 Open Problems Addressed

This paper resolves OPEN-SS-11 (Uniqueness of SU(3) operator mapping from tetrahedral cage geometry), registered 29 March 2026 in the CPP Research Frontier.

2 Definitions

Definition 2.1 (Colour space)**.** *The* colour space *is $V = \mathbb{C}^3$, with orthonormal basis $\{|r\rangle, |g\rangle, |b\rangle\}$ corresponding to the three base vertices $\{V_1, V_2, V_3\}$ of the qCP tetrahedral cage.*
Definition 2.2 (Generator space)**.** *The* generator space *is*

$$\mathcal{H}_0^3 \;=\; \{X \in M_3(\mathbb{C}) \;:\; X = X^\dagger,\ \mathrm{Tr}(X) = 0\}, \tag{1}$$

the real vector space of traceless Hermitian 3×3 matrices.
Definition 2.3 ($\mathfrak{su}(3)$)**.** *The Lie algebra $\mathfrak{su}(3)$ is $\mathcal{H}_0^3$ equipped with the bracket*

$$[X, Y] \;=\; i(XY - YX). \tag{2}$$

Remark 2.4. *The factor of i in (2) ensures the bracket of two Hermitian matrices is Hermitian: $(i[X,Y])^\dagger = -i[Y^\dagger, X^\dagger] = -i[Y,X] = i[X,Y]$. The tracelessness is preserved because $\mathrm{Tr}([X,Y]) = 0$ for any matrices X, Y.*

3 The Uniqueness Theorem

Lemma 3.1 (Dimension of the generator space)**.** $\dim_{\mathbb{R}}(\mathcal{H}_0^3) = 8$.

Proof. A general Hermitian 3×3 matrix has 3 real diagonal entries and 3 complex upper-triangular entries (the lower triangle is determined by $X = X^\dagger$), giving $3 + 2 \times 3 = 9$ real parameters. The tracelessness condition $\mathrm{Tr}(X) = 0$ removes one parameter: $9 - 1 = 8$. □

Lemma 3.2 (CPP operators are a basis)**.** *The eight CPP tetrahedral hopping operators* $\{T^1, \ldots, T^8\}$ *defined in SS-1b (Abshier and Grok, 2026b) are linearly independent over* $\mathbb{R}$ *and therefore form a basis for* $\mathcal{H}_0^3$*.*

Proof. Each T^a is traceless and Hermitian by construction (SS-1b, verified to $< 10^{-16}$). The Gram matrix $G_{ab} = \mathrm{Tr}(T^a T^b)/(1/4)$ has rank 8 and determinant $\det(G) = 3.9 \times 10^{-3} \neq 0$ (numerical verification; see Appendix A). Therefore the eight operators are linearly independent. Since $\dim(\mathcal{H}_0^3) = 8$ (Lemma 3.1), they form a basis. □

Theorem 3.3 (Uniqueness of SU(3))**.** *The Lie algebra generated by the CPP tetrahedral hopping operators is* $\mathfrak{su}(3)$*, and no other Lie algebra is consistent with the tetrahedral cage geometry. Specifically:*

(i) Any set of 8 *linearly independent traceless Hermitian* 3×3 *matrices generates* $\mathfrak{su}(3)$ *under the bracket* (2)*.*

(ii) The CPP operators are such a set (Lemma 3.2).

(iii) Therefore the CPP operators generate $\mathfrak{su}(3)$ *necessarily.*

Proof. By Lemma 3.2, the CPP operators $\{T^a\}$ form a basis for $\mathcal{H}_0^3$. By Definition 2.3, $\mathcal{H}_0^3$ with the bracket (2) *is* $\mathfrak{su}(3)$. A basis for a Lie algebra generates the full algebra under the bracket. Therefore $\{T^a\}$ generates $\mathfrak{su}(3)$.

For uniqueness: suppose an alternative set of 8 operators $\{S^a\} \subset \mathcal{H}_0^3$ were proposed. If they are linearly independent, they are another basis for the same 8-dimensional space $\mathcal{H}_0^3$, and generate the same algebra $\mathfrak{su}(3)$ (possibly with different structure constants $\tilde{f}^{abc}$, corresponding to a different basis convention). If they are linearly dependent (rank < 8), they span a proper subspace and generate a proper subalgebra of $\mathfrak{su}(3)$ — but $\mathfrak{su}(3)$ is *simple* (it has no proper ideals), so its only proper subalgebras have dimension ≤ 3 (isomorphic to $\mathfrak{su}(2)$, $\mathfrak{u}(1)$, or their sums). An alternative set of 8 operators that generates a Lie algebra different from $\mathfrak{su}(3)$ would require dim > 8 or the matrices to be non-Hermitian or to have nonzero trace — all of which violate the cage constraints.

Therefore: **any** set of 8 independent generators satisfying the cage constraints (traceless, Hermitian, 3×3) generates $\mathfrak{su}(3)$, and only $\mathfrak{su}(3)$. □ □

Remark 3.4 (The role of C_3 symmetry)**.** *The* C_3 *rotational symmetry* $V_1 \to V_2 \to V_3 \to V_1$ *acts on* $\mathcal{H}_0^3$ *by conjugation:* $T^a \mapsto P T^a P^{-1}$*, where* P *is the cyclic permutation matrix. This maps the generators into linear combinations of each other (verified numerically; see Appendix A), confirming that* C_3 *is an inner automorphism of* $\mathfrak{su}(3)$*. The* C_3 *symmetry selects the* basis *(the Gell-Mann convention) but does not alter the* algebra*. With or without* C_3*, the algebra is* $\mathfrak{su}(3)$*.*

4 Corollaries

Corollary 4.1 (Gauge group from vertex count)**.** *The gauge group of the strong sector is determined entirely by the number of base vertices* N *of the qCP cage. For a cage base with* N *vertices, the gauge group is SU(*N*).*

N	$\dim(\mathfrak{su}(N))$	*Gauge group*	*CPP sector*
2	*3*	*SU(2)*	*Electroweak (single edge $V_1 \leftrightarrow V_2$)*
3	***8***	***SU(3)***	***Strong (tetrahedral base)***
4	*15*	*SU(4)*	*Not realised (would require square base)*

The 600-cell is composed of tetrahedral cells ($N = 3$ base vertices plus one apex). Therefore SU(3) is the unique gauge group of the strong sector. SU(2) arises when only one edge of the base triangle is activated (the electroweak sector; EW-5 (Abshier and Grok, 2026c)), confirming that SU(2) and SU(3) differ only in the number of tetrahedral edges engaged.

Corollary 4.2 (No exotic gauge groups)**.** *No exotic gauge group — SO(8),* Sp(4)*, G_2, or any other — can arise from three colour states. The only* 8*-dimensional Lie algebra that acts faithfully on $\mathbb{C}^3$ via traceless Hermitian matrices is* $\mathfrak{su}(3)$*.*

Proof. Any faithful action of a Lie algebra $\mathfrak{g}$ on $\mathbb{C}^3$ via traceless Hermitian matrices embeds $\mathfrak{g}$ as a subalgebra of $\mathcal{H}_0^3 = \mathfrak{su}(3)$. If $\dim(\mathfrak{g}) = 8 = \dim(\mathfrak{su}(3))$, the embedding is surjective and $\mathfrak{g} \cong \mathfrak{su}(3)$. If $\dim(\mathfrak{g}) < 8$, it is a proper subalgebra (not a candidate for the full gauge group). If $\dim(\mathfrak{g}) > 8$, the embedding into the 8-dimensional space is impossible.

Specific exclusions:

- $\mathfrak{so}(8)$ has $\dim = 28 > 8$: cannot embed.
- $\mathfrak{sp}(4)$ has $\dim = 10 > 8$: cannot embed.
- $\mathfrak{g}_2$ has $\dim = 14 > 8$: cannot embed.
- $\mathfrak{su}(2) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1) \oplus \mathfrak{u}(1)$ has $\dim = 8$, but the faithful representation of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ requires $\mathbb{C}^4$ (not $\mathbb{C}^3$). □

□

Corollary 4.3 (Why 3 colours, not 2 or 4)**.** *The 600-cell consists exclusively of tetrahedral cells. Each tetrahedron has* 4 *vertices:* 1 *apex (V_4, the qCP site) and* 3 *base vertices ($\{V_1, V_2, V_3\}$, the colour states). The number* 3 *is not a choice — it is the vertex count of the face opposite the apex in a tetrahedron. Since the 600-cell admits no non-tetrahedral cell types (Coxeter, 1973), CPP cannot produce a gauge group other than SU(3) for the strong sector or SU(2) for the electroweak sector.*

5 Discussion

5.1 What the uniqueness theorem means for CPP

SS-1 Theorem 1 proved that the CPP cage operators reproduce the Gell-Mann matrices. A sceptic could respond: "So what? Perhaps a clever choice of operators on any 3-vertex graph would give SU(3). The cage didn't force it — you engineered it."

The uniqueness theorem closes this objection. The cage provides *exactly* $3^2 - 1 = 8$ independent operator degrees of freedom (3 edges × 2 hopping modes + 2 diagonal phases). *Any* basis for these degrees of freedom generates $\mathfrak{su}(3)$. The only "engineering" is the vertex count $N = 3$, which is determined by the 600-cell's tetrahedral cell geometry. Once $N = 3$ is established, SU(3) is inevitable.

5.2 The $8 = 3 \times 2 + 2$ decomposition

The count of 8 generators decomposes as:

$$\underbrace{3 \text{ edges} \times 2 \text{ (real + imaginary)}}_{\text{6 colour-changing}} + \underbrace{2 \text{ diagonal phases}}_{\text{2 colour-neutral}} = 8 = \dim(\mathfrak{su}(3)). \tag{3}$$

This is the same count as Grok's layer-depth argument ($1 + 3 + 4 = 8$; SS-1 Remark 3.1 (Abshier and Grok, 2026a)). Both routes confirm that the dimension 8 is a geometric consequence of the tetrahedron, not a numerical coincidence.

5.3 Connection to the electroweak sector

When only one edge of the base triangle is activated (say $V_1 \leftrightarrow V_2$), the operator count is $1 \times 2 + 1 = 3 = \dim(\mathfrak{su}(2))$, recovering the weak isospin algebra. The two fundamental forces of the Standard Model — SU(3) for the strong force and SU(2) for the weak force — differ only in how many tetrahedral edges participate. The uniqueness theorem confirms this is not a coincidence: SU(N) is the unique algebra of N vertices, and the tetrahedral cage activates either $N = 2$ (one edge) or $N = 3$ (all edges).

6 Conclusion

The SU(3) Lie algebra is the *unique* Lie algebra consistent with the CPP tetrahedral cage geometry. The proof requires only three ingredients: (i) the dimension formula $\dim(\mathcal{H}_0^N) = N^2 - 1$, (ii) the definition of $\mathfrak{su}(N)$ as the bracket algebra of traceless Hermitian matrices, and (iii) the linear independence of the CPP operators. No exotic gauge group can arise from three colour states. The number 3 is itself forced by the 600-cell's tetrahedral cell geometry.

SS-1 Theorem 1 is thereby elevated from a *possibility* result to a *necessity* result: the tetrahedral cage does not merely *accommodate* SU(3) — it *requires* it.

6.1 Problem Status After This Paper

- OPEN-SS-11 (SU(3) operator uniqueness): OPEN → **THEO** (proved in Theorem 3.3).
- SS-1 Theorem 1: elevated from possibility to necessity.

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OSF project: `https://osf.io/9dfya/`
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A Numerical Verification

All computations performed in Python 3 with NumPy. Source code available at `CPP/series_strong/notebooks/SS-3_su3_uniqueness.py`.

A.1 Linear independence

The Gram matrix $G_{ab} = 4\ \mathrm{Tr}(T^a T^b)$ has:

- Rank: 8 (full rank).
- Determinant: $\det(G) = 3.906 \times 10^{-3} \neq 0$.

The factor 4 normalises so that $G_{ab} = \delta_{ab}$ for the standard Gell-Mann convention (which uses $\mathrm{Tr}(T^a T^b) = \delta_{ab}/2$; our $T^a = \lambda^a/2$ gives $\mathrm{Tr}(T^a T^b) = \delta_{ab}/4$ for $a \neq 8$, with T^8 having a different normalisation).

A.2 Commutation closure

For all $8 \times 8 = 64$ pairs (a, b):

$$\max_{a,b} \left\| [T^a, T^b] - i \sum_c f^{abc} T^c \right\| < 1.1 \times 10^{-16} \tag{4}$$

where $f^{abc} = 2\ \mathrm{Tr}([T^a, T^b]T^c/i)$ (trace extraction). This confirms closure to machine precision.

A.3 C_3 action

The permutation matrix P ($V_1 \to V_2 \to V_3 \to V_1$) satisfies $P^3 = I$. Under conjugation $T^a \mapsto PT^aP^{-1}$:

- The six off-diagonal operators permute among themselves (edges cycle: $12 \to 23 \to 31$).
- The two diagonal operators mix: $T^3 \mapsto -\frac{1}{2}T^3 + \frac{\sqrt{3}}{2}T^8$ and $T^8 \mapsto -\frac{\sqrt{3}}{2}T^3 - \frac{1}{2}T^8$ (a rotation by $2\pi/3$ in the Cartan subalgebra).

The C_3 action maps $\mathcal{H}_0^3$ into itself, confirming it is an inner automorphism of $\mathfrak{su}(3)$.

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